

Internal Research Report: Do Temporal Network Signals Predict How Scientific Concepts Spread Across Disciplines?

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1 Run overview

This report is the chronological record of a five-iteration research run investigating whether early structural signals in a concept’s topic co-occurrence network anticipate later cross-field breadth, beyond simple measures of volume and reach. The run used the OpenAlex bulk snapshot (476,196,327 works; 2026-09-23) and identified 12,499 concepts, tracked 27,393 adoption episodes across 26 fields, and screened 53 early network indicators against a five-feature popularity baseline.

Twenty artifacts were commissioned across five iterations; sixteen completed and four failed. The run progressed through several hypothesis shifts: from citation naturalisation (iteration 1, null) to gateway-field retention (iteration 2, disconfirmed at scale) to retained-frontier relatedness (iteration 3, partial) to early ego-network openness (iterations 4–5, lead). Throughout, the headline claim narrowed and confidence decreased.

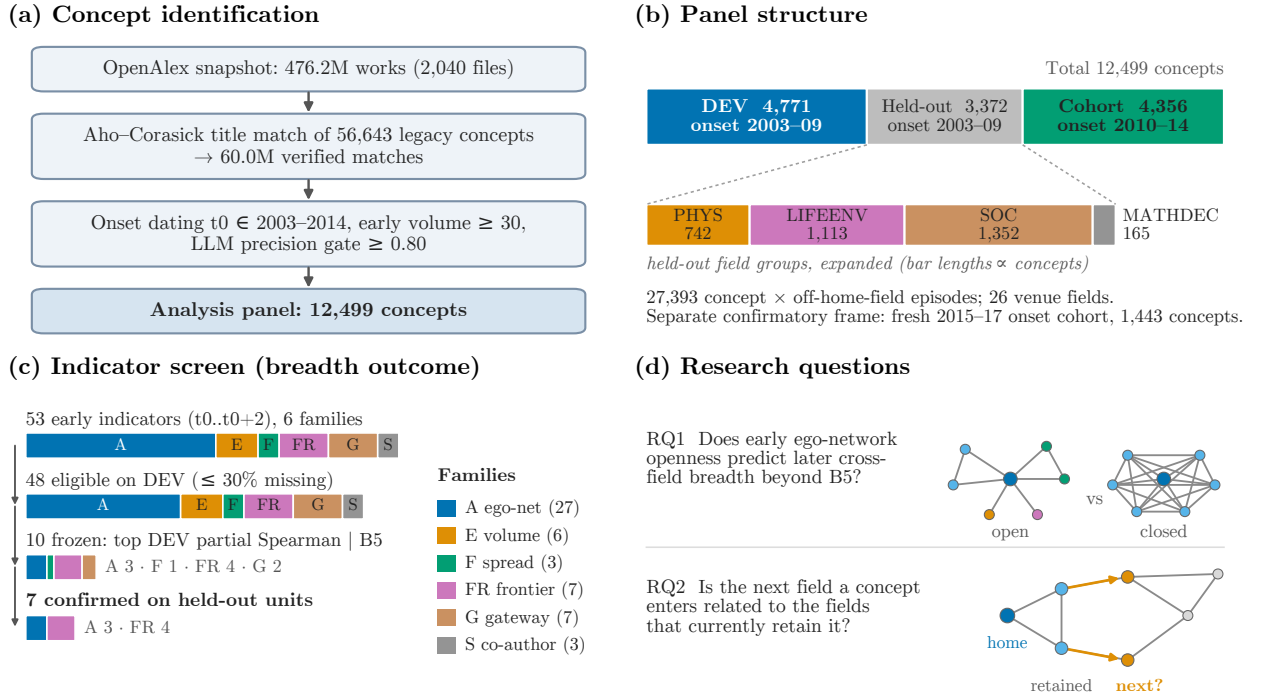


Figure 1: Overview of the study design. (a) Concept identification: 56,643 legacy OpenAlex concepts are matched by title (Aho-Corasick) against a 476.2M-work OpenAlex snapshot, giving 60.0M verified matches; concepts are onset-dated ($t_0 \in 2003-2014$), required to have early volume ≥ 30 and to pass an LLM precision gate ≥ 0.80 , yielding an analysis panel of 12,499 concepts. (b) Panel structure, bar lengths proportional to concept counts: DEV (blue; CS, Eng, BGM and Med homes, onset 2003–09) 4,771, held-out field groups (grey; onset 2003–09) 3,372, and the 2010–14 onset cohort (green) 4,356. The expanded lower bar splits the held-out block into PHYS 742, LIFEENV 1,113, SOC 1,352 and MATHDEC 165. The panel contains 27,393 concept $\times$ off-home-field episodes over 26 venue fields; a separate fresh 2015–17 onset cohort of 1,443 concepts serves as a confirmatory frame. (c) Indicator screen for the breadth outcome (O2r_m50). Bars are coloured by indicator family with lengths proportional to indicator counts: 53 early indicators computed over $t_0..t_0+2$; 48 eligible on DEV; the 10 frozen on DEV by partial Spearman correlation given the five-feature baseline B5; and the 7 confirmed on held-out units. (d) The two research questions as schematic illustrations.

2 Iteration 1: Wide screen of rival spread signals

2.1 Strategy and rationale

The run’s first iteration was designed as a wide screen of five rival mechanisms for why new concepts become broadly integrated. The prior review had computed a reliability of approximately 0.32 for the initial candidate A_h^* (background-adjusted lineage naturalisation gap). Rather than deepening this fragile signal, iteration 1 raced five candidates on one frozen DEV panel: (1) lineage naturalisation A_h^* , (2) unconnected co-author groups (candidate S), (3) co-occurrence structural diversity D , (4) frequency-free selectivity F , and (5) gateway-field landing G . Five artifacts were commissioned.

2.2 Experiment 1: Does citing a concept ‘as your own’ predict its spread?

This artifact screened candidate A_h^* on the frozen P78 dev panel (48 concepts: Biochem 13, CS 21, Engineering 3, Medicine 11).

Result. A_h^* does *not* survive the pre-registered rule:

- LOGO $\Delta\rho$ for O2r over B5 = -0.006 , 90% concept-bootstrap CI $[-0.034, 0.017]$; $\rho_{B5} = 0.834$. FAIL.
- Positive left-out groups: 0 of 4. FAIL.
- Split-half reliability (Spearman–Brown) = 0.58. FAIL (bar 0.6).
- $|\rho|$ with log early volume / early growth = 0.14/0.18. PASS.

Background homophily finding (M1). Among 48 concepts, 100% have a positive background log-odds ratio: every field’s citing papers preferentially cite their own field above chance. R^2 of the raw concept-lineage log-odds ratio on the background log-odds ratio is 0.66 (90% CI $[0.39, 0.83]$). Two-thirds of between-concept variance in raw lineage autonomy is general disciplinary homophily, not concept-specific integration.

Variance decomposition. REML: $\tau_c = 0.29$, $\tau_{cj} = 0.65$. PyMC NUTS check passes (Spearman 0.9996 with REML).

Data deviation. The shared OpenAlex credit pool ran dry (139 own credits spent). Yearly counts came from OpenAlex S0; field labels, citation lineage and background references came from free Semantic Scholar data.

Table 1: Experiment 1 screen result: candidate A_h^* clause-by-clause (iteration 1, `gen.art.experiment_1`).

Clause	Value	Pass
$\Delta\rho \geq 0.10$, CI low > 0	$-0.006 [-0.034, 0.017]$	No
$\geq 3/4$ groups positive	0 of 4	No
Split-half $r_{SB} \geq 0.6$	0.583	No
$ \rho $ with vol/growth ≤ 0.6	0.14 / 0.18	Yes

2.3 Experiment 3: Do diverse topic ties predict concept spread?

This artifact screened two co-occurrence emergence indicators (D_{ratio} and F_{res}) on 47 DEV concepts under shared protocol S0.

Result. Neither candidate survives:

- D_{ratio} : $\Delta\rho = +0.006$, CI90 $[-0.092, 0.135]$, 3/4 groups positive, $r_{SB} = 0.83$.
- F_{res} : $\Delta\rho = -0.060$, CI90 $[-0.158, 0.014]$, 1/4 groups positive, $r_{SB} = 0.44$.

Exploratory partial association. The out-of-group partial ρ of D_{ratio} given B5 was 0.335 [0.02, 0.65], permutation $p = 0.037$; $\Delta\rho$ is near its ceiling because B5 is already strong ($\rho_{B5} = 0.770$).

Portability. D_{ratio} , D_{rare} , participation and NOV_{res} are associated with O2r in all 4 groups (pooled ρ 0.45–0.63; within-group minima as low as 0.12 for participation in CS, 0.27 for NOV_{res} in CS, 0.33 for D_{ratio} in Eng), but are redundant under $\Delta\rho$. Degree, strength and new-edge growth are CS-only (a negative result).

Table 2: Experiment 3 screen result (iteration 1, gen_art_experiment_3).

Indicator	$\Delta\rho$	CI90	Groups+	r_{SB}	Survives
D_{ratio}	+0.006	$[-0.092, 0.135]$	3/4	0.83	No
F_{res}	-0.060	$[-0.158, 0.014]$	1/4	0.44	No

2.4 Experiment 4: Where a concept lands early vs how broadly it spreads

This artifact screened gateway landing G on 46 DEV concepts and produced the authoritative S0 outcome tables.

Result. G does not survive the pre-registered rule: $\Delta\rho = +0.033$, CI90 $[-0.095, 0.168]$, positive in 2/4 groups, $r_{SB} = 0.92$. However, B5 reaches only $\rho = 0.327$ on these 34 concepts with an outcome window (per group: CS 0.10, Eng 0.86, BGM 0.65, Med 0.57), much weaker than in Experiments 1 and 3. So the ceiling argument does not apply to G .

Field-level retention lead. Adding the adopting field’s eigenvector gateway centrality raised retention AUC from 0.705 to 0.808 (+0.103, 95% CI [0.034, 0.167]) on 80 episodes. This was the only lead from iteration 1, carried forward for testing.

Next-field entry. Relatedness density AUC 0.61 beats the permutation null ($p = 0.023$) but loses to log field size (0.74).

Table 3: Experiment 4 screen result (iteration 1, gen_art_experiment_4).

Indicator	$\Delta\rho$ (O2r)	CI90	Groups+	r_{SB}
G (gateway landing)	+0.033	$[-0.095, 0.168]$	2/4	0.92

2.5 Failed artifacts

Two iteration-1 artifacts failed:

- **gen_art_dataset_1** (Sealed test set of new science concepts): stalled, never completed. The outcome-blind held-out frame was not built.
- **gen_art_experiment_2** (Do independent author groups predict spread?): candidate S, testing Cheng et al.’s co-author independent-groups hypothesis, stalled and never ran. Candidate S was therefore untested, not refuted, at this stage.

2.6 Iteration 1 dead ends

1. A_h^* naturalisation gap: null ($\Delta\rho = -0.006$, 0/4 groups).
2. Co-occurrence diversity D_{ratio} : null ($\Delta\rho = +0.006$), though the exploratory partial association is marginal.
3. Frequency-free selectivity F_{res} : null ($\Delta\rho = -0.060$), unreliable ($r_{SB} = 0.44$).
4. Gateway landing G : null on concept-level breadth ($\Delta\rho = +0.033$), but the field-level retention lead (+0.103) is carried forward.
5. CS-only co-occurrence growth indicators (degree, strength, new-edge-rate): do not generalise beyond CS.

2.7 Iteration 1 review and hypothesis update

The review scored 3 (blocking) with soundness 1, citing conclusions that contradicted the run’s own evidence (e.g., a ceiling argument applied to Exp4 where $\rho_{B5} = 0.327$). The hypothesis update moved from A_h^* to the field-level gateway-retention lead ($\Delta\text{AUC} +0.103$, dev only, 80 episodes), with the unit of analysis shifting to concept $\times$ field adoption episodes. The move was **deepen**: build the held-out frame from the free snapshot, test gateway retention at scale, and address RQ2 (next-field entry and trajectories).

3 Iteration 2: Do hub fields keep new concepts?

3.1 Strategy and rationale

Iteration 2 moved the budget from more metrics to more samples. Five artifacts were commissioned: (1) Experiment 5, a decisive held-out test of gateway retention on a zero-credit 12,499-concept panel; (2) Experiment 6, testing next-field entry and trajectories; (3) Evaluation 1, stress-testing the iteration-1 lead; (4) Dataset 2, external-recognition ground truth; (5) Research 1, related-work positioning.

3.2 Experiment 5: Held-out test of gateway-field retention

One zero-credit scan of the full OpenAlex S3 snapshot built the authoritative S1 frame: 12,499 concepts (DEV 4,771; held-out PHYS 742, LIFEENV 1,113, SOC 1,352, MATHDEC 165; cohort 4,356) and 27,393 episodes.

H1 result: DISCONFIRMED. Held-out $\Delta\text{AUC} = -0.00001$ $[-0.0006, +0.0003]$; DerSimonian–Laird pooled -0.00004 ($I^2 = 0$). Minimum detectable ΔAUC : 0.004.

Baseline ladder. Gateway’s DEV signal ($+0.0019$ over the iteration-1 base) vanishes once the leave-concept-out field propensity $P_j(-c)$ is added, and reverses on held-out (-0.0016). Gateway alone has AUC 0.605 on DEV vs 0.506 on held-out (0.41 in SOC). Gateway is a domain-specific proxy for “fields that keep things,” not a position-dependent causal factor.

H3 (concept-level gateway landing). Held-out partial ρ : $G = 0.030$, $G_A = 0.026$, $G_{\text{btw}} = 0.046$ (Holm $p = 0.0045$); DL-pooled $G = 0.068$ $[0.029, 0.107]$. The concept-bootstrap CI of pooled G is $[-0.006, 0.065]$, which includes zero. DEV-to-held-out shrinkage is from 0.138 to 0.030.

Table 4: Experiment 5 panel composition (iteration 2, `gen_art_experiment_5`).

Split	Concepts	Episodes
DEV (CS/Eng/BGM/Med, onset 2003–09)	4,771	9,079
Cohort (onset 2010–14, all fields)	4,356	9,799
Held-out Physical Sciences	742	1,662
Held-out Life & Environment	1,113	3,099
Held-out Social Sciences	1,352	3,320
Held-out Math & Decision Sci.	165	434
Total	12,499	27,393

3.3 Experiment 6: Where new scientific concepts spread next

Tested the retained-frontier claim on 653 newborn concepts (1,865 episodes): does relatedness to fields currently retaining a concept predict which field it enters next?

H2 result: CONFIRMED by the frozen rule. Held-out LR = 71.7 ($p = 2 \times 10^{-17}$), $d_0 = 0.302$ $[0.240, 0.369]$; DL pooled $d_0 = 0.284$ $[0.216, 0.352]$, $I^2 = 0$. Label-permutation $p = 0.001$, rewired-backbone $p = 0.015$.

Gateway weighting adds nothing. M3 vs M1 permutation $p = 0.17$ (held-out), 0.31 (DEV).

Trajectories. DTW k -medoids $k = 2$ stable (bootstrap ARI 1.0): volume-matched “integrating” vs “localised” classes (held-out independent recluster ARI 0.54; localised class dominated by Medicine homes). The HMM (6 states) gives HMM-vs-DTW ARI of 0.094, a robustness failure of the “two stable classes” claim.

Ordering. First retained gateway field precedes entropy take-off in 66% of broad concepts (sign $p = 0.003$) vs 57% for peripheral fields (McNemar $p = 0.09$). However, the lead-lag regression shows negative coefficients, a significant pre-trend ($\text{ev}_{-3} = -0.072$, $p = 0.0002$), and on DEV the reverse path is significant ($b = 0.232$, $p = 0.006$). The ordering result was later reclassified as MIXED.

3.4 Evaluation 1: Stress-testing the gateway-field retention lead

Result: gateway retention FAILS. Union panel (362 de-duplicated episodes, 54 concepts): $\Delta\text{AUC} = +0.001$ $[-0.012, 0.012]$; DL pooled $+0.0015$ ($I^2 = 0$). The shuffled- R placebo’s 95th percentile (0.130) exceeds the original $+0.103$, so the iteration-1 lead cannot be certified as above chance on 80 episodes.

O1 label-coverage artefact. All 8 G -variant O1 gains ($+0.05.. +0.15$) are label-coverage artefacts: $G +0.072 \rightarrow +0.002$ once label coverage is controlled.

3.5 Dataset 2: External recognition lookup table

Built external-recognition dates (outcome O5) for 65,026 OpenAlex legacy concepts from MeSH (20,872 concepts), English Wikipedia (6,540 exact), Wikidata P571/P575 (1,425), ACM CCS, MSC, PACS/PhySH, and curated breakthrough lists. O5 was later evaluated and found unrelated to publication outcomes: pooled ρ with O2r_{m50} is 0.014 $[-0.045, 0.073]$; 67% of concepts are recognised at or before t_0 .

3.6 Research 1: How our results compare with related papers

Positioning study for the Applied Network Science paper. Collected 22 citable ANS papers with relation lines. Identified comparison numbers: Guevara (2016) field-entry AUC 0.896/0.715/0.682 for individuals/organisations/countries; no published retention AUC benchmark.

3.7 Iteration 2 dead ends

1. Gateway-field retention: DISCONFIRMED at scale ($\Delta\text{AUC} = -0.00001$).
2. H3 concept-level gateway landing: effect about 0.03 partial ρ , a quarter of its DEV value; pooled CI includes zero.
3. Ordering: MIXED (negative lead-lag, pre-trend, reverse path significant on DEV).
4. Two trajectory classes: DTW $k = 2$ not reproduced by HMM (ARI 0.094); localised class dominated by Medicine.
5. G -variant O1 gains: label-coverage artefacts.
6. Rescue and relay: not supported on held-out data.

3.8 Iteration 2 review and hypothesis update

Review scored 3 (blocking), soundness 1. The hypothesis moved from gateway centrality (closed) to the retained-frontier lead from Experiment 6 ($d_0 = +0.281$). The decisive test for iteration 3: beat the conventional RCA-thresholded Hidalgo density on an independent frame.

4 Iteration 3: Decisive tests on the large panel

4.1 Strategy and rationale

Five artifacts were commissioned: (1) Experiment 7, the decisive retained-frontier test against the RCA-thresholded rival on an independent frame; (2) Experiment 8, the RQ1 held-out indicator

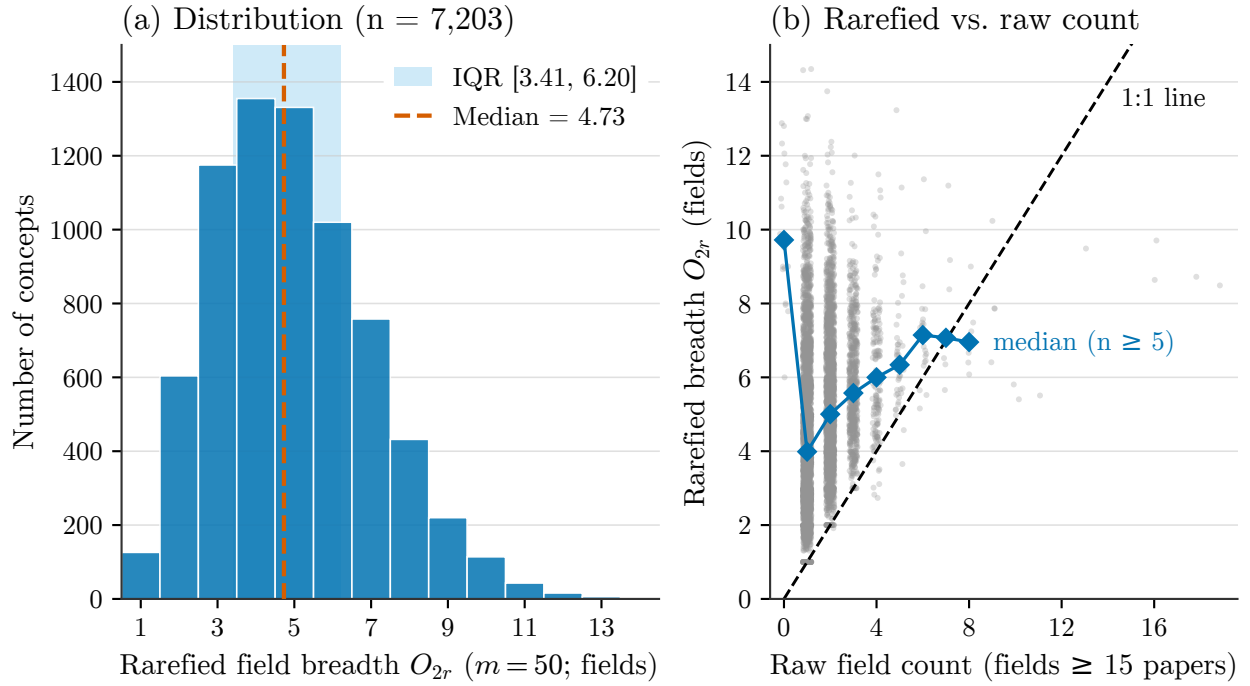


Figure 2: Distribution of the primary outcome, rarefied field breadth (O_{2r} , $m = 50$), over the 7,203 of 12,499 frame concepts that have at least 50 papers in the $t_0+6..t_0+8$ outcome window. (a) Histogram of O_{2r} ; the dashed line marks the median (4.73) and the light band the IQR [3.41, 6.20]. (b) O_{2r} against the raw field count.

screen (53 indicators, 6 families); (3) a planned RQ2 trajectories artifact (gen_art_experiment_9) that failed; (4) Evaluation 2, auditing the record; (5) Research 2, prior art and venue check.

4.2 Experiment 7: Do concepts spread from fields that keep them?

Independent frame: 11,841 concepts (Exp5 minus every Exp6 concept). DEV 4,486 used for code; held-out scored once.

Result: FRONTIER = PARTIAL. Held-out pooled $d_0 = 0.322$ [0.291, 0.355], LR(R3 vs R2) = 325.8 ($p < 10^{-70}$). Positive in PHYS (0.15), LIFEENV (0.40), SOC (0.30); MATHDEC (0.07, underpowered). DL over 4 groups: 0.243 [0.118, 0.368], $I^2 = 0.92$. Retained-label permutation $p = 0.001$, rewire $p = 0.004$.

Volume-matched contrast: null. The pre-declared volume-matched contrast (retained vs entered-not-retained fields in the same volume cell) is -0.028 $[-0.105, 0.046]$ on held-out (DEV: -0.008). Frozen verdict: FRONTIER = PARTIAL (“persistence confounded with volume”).

Proximity dependence. Under Hidalgo min-conditional-probability proximity, $d_0 = -0.021$ ($p = 0.012$). The effect is backbone-specific.

Dose by retention age. Held-out: 2 years +0.098, 3 years +0.075, ≥ 4 years +0.304. Not monotone (held-out monotone flag: false).

Table 5: Experiment 7 field-entry results on the independent frame (iteration 3, gen_art_experiment_7). Held-out pooled.

Model	d_0	95% CI	LR vs prior	Verdict
R0 (home, size, density, gateway)	—	—	—	—
R1 ($+D_{\text{rca},1y}$)	—	—	—	—
R2 ($+D_{\text{vol}}$)	—	—	—	—
R3 ($+d_0$ retained)	0.322	[0.291, 0.355]	325.8	Confirmed
S.strict (all rivals)	0.304	[0.268, 0.336]	—	Survives
Volume-matched contrast	-0.028	$[-0.105, 0.046]$	—	Null
Min-cp proximity	-0.021	—	—	Reverses

4.3 Experiment 8: Which early network signals travel across fields

The RQ1 held-out deliverable: 53 indicators in 6 families (A co-occurrence ego-network 27, E popularity 6, F disciplinary spread 3, FR retained frontier 7, G gateway landing 7, S co-author 3) over $t_0.t_0 + 2$, plus B5 baseline. Outcomes: O1c/O1b uptake, O2r breadth, O3 transience, O4 citation growth, O5 external recognition.

Breadth result. 7 of 10 frozen indicators confirmed with 6/6 unit sign agreement for O2r_{m50}; 8 of 10 for O2r_{resid}.

Learned models. ElasticNet combining all indicators adds Spearman +0.059 [+0.046, +0.073] over B5 on held-out; EBM adds +0.052 [+0.037, +0.067].

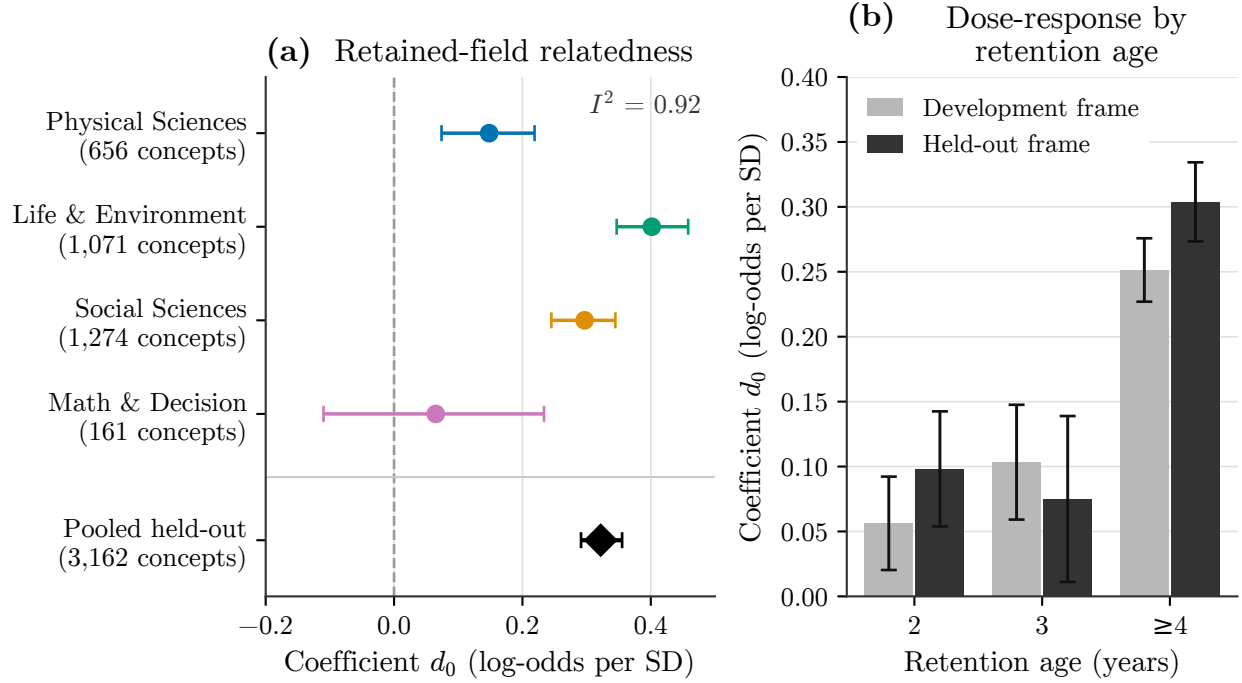


Figure 3: Retained-field relatedness predicts the next field a concept enters. (a) Conditional-logit coefficient d_0 on the held-out groups of the independent frame. Coloured circles are per-group estimates with 95% concept-bootstrap CIs. The black diamond is the pooled estimate, $d_0 = 0.322$ [0.291, 0.355]. (b) d_0 split by retention age (2, 3, ≥ 4 years).

Table 6: Held-out indicator screen: the 10 frozen indicators for rarefied breadth ($O2r_{m50}$), tested on held-out groups. DerSimonian–Laird pooled estimates with 95% CI (iteration 3, `gen.art.experiment_8`).

Indicator	Family	Pooled PSP	95% CI	I^2	Sign	Confirmed
M0_density_end	FR	+0.375	[+0.279, +0.462]	0.74	6/6	Yes
D_vol_end	FR	+0.307	[+0.256, +0.356]	0.10	6/6	Yes
CONTACT_REACH	FR	+0.211	[+0.161, +0.261]	0.00	6/6	Yes
n_{comm} (W3)	A	+0.167	[+0.063, +0.267]	0.78	6/6	Yes
NOV_{res}	A	+0.151	[+0.044, +0.255]	0.75	6/6	Yes
RETENTION_RATIO	FR	−0.114	[−0.160, −0.067]	0.00	6/6	Yes
ego_density (W3)	A	−0.102	[−0.151, −0.053]	0.00	6/6	Yes
Rao–Stirling	A	−0.072	[−0.153, +0.010]	0.44	5/6	No
G_{btw}	G	+0.056	[−0.006, +0.118]	0.33	6/6	No
log offhome vol.	F	−0.089	[−0.171, −0.007]	0.63	5/6	No

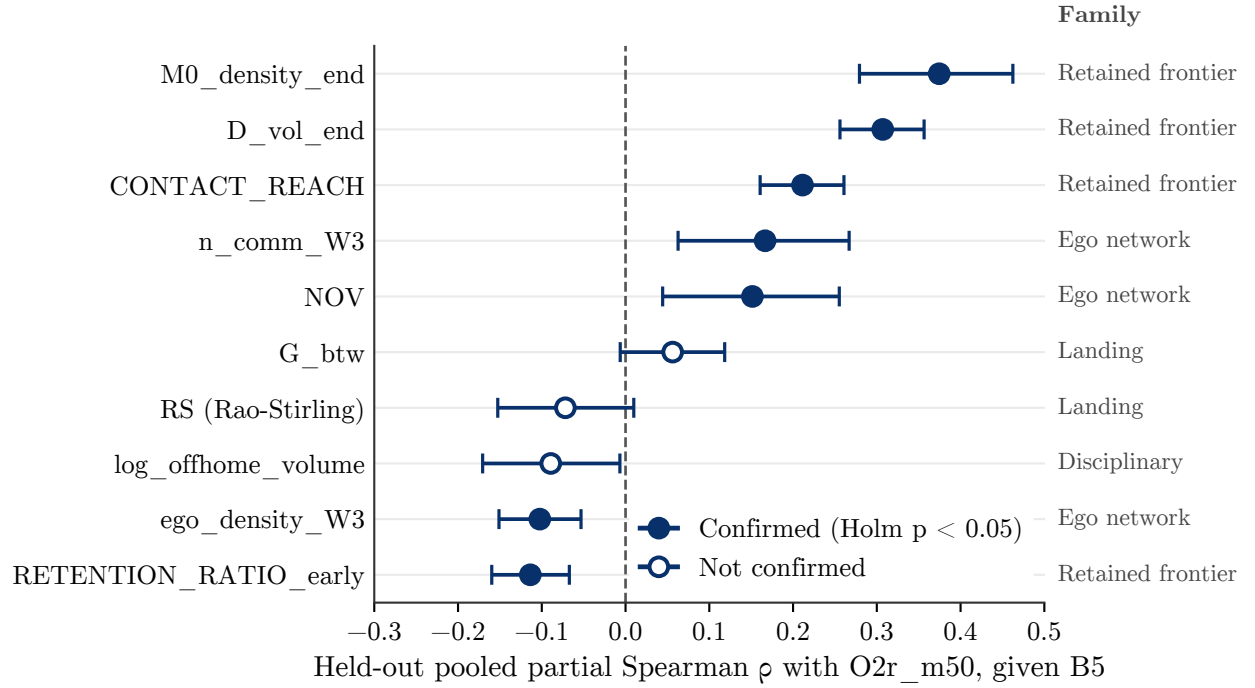


Figure 4: Held-out screen of the 10 frozen indicators for rarefied cross-field breadth (O2r_m50). Each row shows the DerSimonian–Laird pooled partial Spearman ρ given B5. Filled circles are confirmed; open circles are not. Seven of the ten are confirmed across two families: retained frontier (FR) and co-occurrence topology (A).

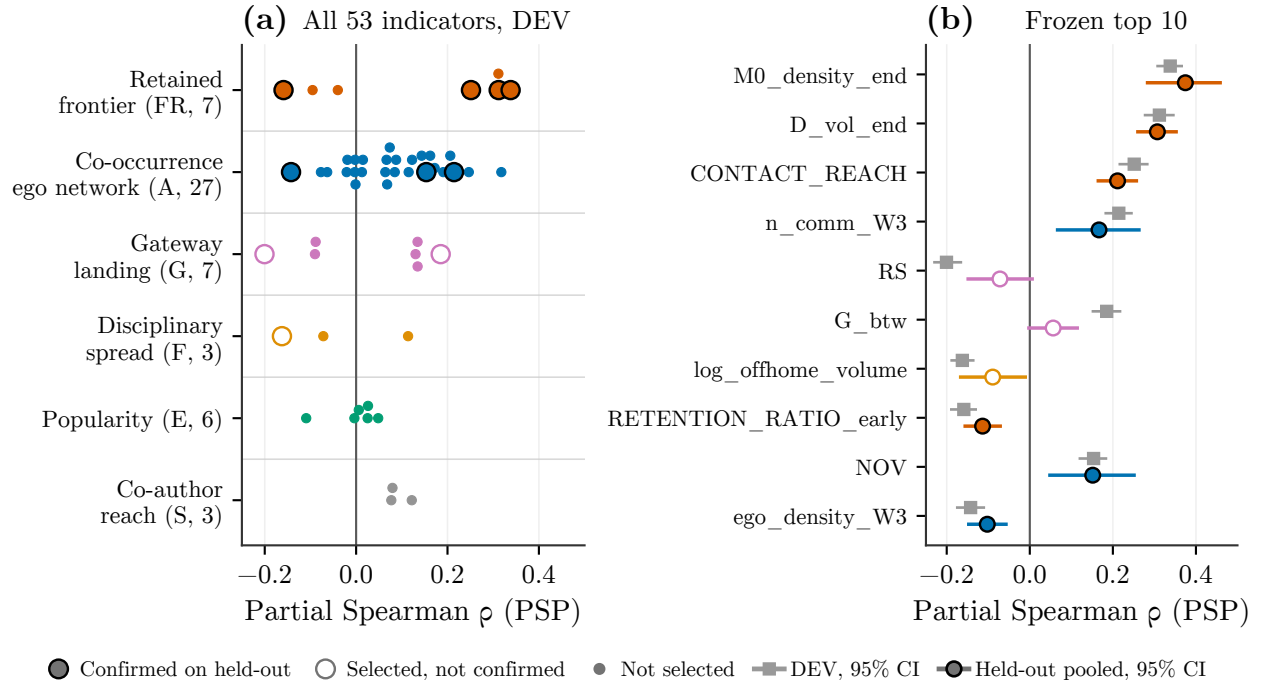


Figure 5: Complete indicator screen for rarefied cross-field breadth (O2r). (a) PSP of all 53 screened indicators on the DEV fields, one row per family. Large filled circles are the 7 confirmed; large open circles are the 3 selected but not confirmed; small dots are the 43 not selected. (b) The frozen top 10 in DEV rank order with held-out pooled estimates.

Pre-onset footprint caveat. M0_density_end and D_vol_end use cumulative field history from 1995 to $t_0 + 2$. Part of their signal therefore predates onset.

Secondary outcomes. O1c: only n_authors_early confirmed (+0.161). O4 citation growth: REL_home -0.114, author_growth +0.065; EBM Spearman 0.188 vs B5 0.015. O3 transience: n_authors_early is the only confirmed indicator; L1-logit AUC +0.093 over B5 at chance (0.506). O5 external recognition: no indicator or model beats B5 + onset year.

4.4 Failed artifact: gen_art_experiment_9

The planned RQ2 trajectories artifact (contact $\times$ retention decomposition, DTW+HMM typology, sequence tests, case studies) was commissioned but never executed: the output-format validation loop failed after 5 retries. This was iteration 3’s entire RQ2 artifact.

4.5 Evaluation 2: Auditing the record before the paper

Audited 246 claims from the record: 224 MATCH, 15 MISLABELLED, 6 MISMATCH, 1 FILE_FLAG_OVERRIDDEN. Produced insert-ready corrections in 14 blocks and record tables. Validated O5: precision 0.86, dates within 1 year 95%, false-negative rate ≥ 0.14 .

4.6 Research 2: Is ‘fields that keep it’ new?

Claim A (next-field entry follows relatedness to retained fields): PARTIALLY ANTICIPATED. No prior study uses a retained-only or duration-weighted density predictor. Claim B (relatedness to dropped fields lowers entry): mechanism anticipated, test NEW. Identified missing rival: persistence-filtered RCA density.

4.7 Iteration 3 review and hypothesis update

Review scored 3 (blocking), soundness 1. Key finding: the Experiment 8 indicator screen confirmed 7 indicators, shifting the headline from field-level retained frontier to concept-level ego-network openness. The hypothesis update moved to **deepen**: confirm the openness index on a fresh 2015–17 cohort, test confounds (concept type, pre-onset footprint, mechanical coupling), and rebuild RQ2 trajectories.

5 Iteration 4: Does openness predict spread? Fresh-cohort test

5.1 Strategy and rationale

Five artifacts were commissioned: (1) Experiment 10, fresh-cohort confirmation of OPEN; (2) Experiment 11, within-concept closure test (incomplete); (3) Experiment 12, RQ2 decomposition and trajectories; (4) Evaluation 3, record repair and boundary study; (5) Research 3, novelty check.

5.2 Experiment 10: Do open-neighbourhood concepts spread? Fresh-cohort test

A single-unseal confirmation on a 2015–2017 onset cohort (after the declared power extension; $n = 1,443$; pre-seal power 0.16) never used in any prior screen. OPEN is the mean of six signed, z-scored ego-network components, with constants frozen on the 12,499 Exp5 concepts.

Result: CONFIRMED but marginal. $\text{OPEN}_{\text{home}}$ partial Spearman with O2r_{m50} is $+0.091$ $[+0.013, +0.171]$ at R2 and $+0.080$ $[+0.001, +0.162]$ at R3. CIs include 0 at R4/R5; DL pool over groups: $+0.083$ $[-0.007, +0.173]$; Holm $p = 0.048$. Predictive gain is negligible: B5 Spearman 0.768 vs 0.770 $(+0.002$ $[-0.003, +0.008])$.

Mechanical coupling. OPEN_{all} : $+0.174$; ALL minus HOME $+0.093$ $[+0.016, +0.169]$. About half of Exp8’s openness signal was coupling. $\text{OPEN}_{\text{sizematch}}$ (corpus-wide papers subsampled to home counts) is in between.

Components. NOV_{res} ($+0.134$) and low edge persistence (-0.112) carry the home-only signal; n_{comm} ($+0.002$) and participation ($+0.050$) are null in the home build.

Table 7: OPEN index control ladder on the 2015–2017 confirmatory cohort. PSP = partial Spearman with O2r_{m50} , 95% CI (iteration 4, gen_art_experiment_10).

Build	R0	R1	R2	R3	R4	R5	n
$\text{OPEN}_{\text{home}}$	$+0.12$	$+0.10$	$+0.09$	$+0.08$	$+0.07$	$+0.06$	573
OPEN_{all}	$+0.21$	$+0.18$	$+0.17$	$+0.17$	$+0.15$	$+0.14$	630
$\text{OPEN}_{\text{size}}$	$+0.18$	$+0.15$	$+0.15$	$+0.14$	$+0.12$	$+0.11$	591

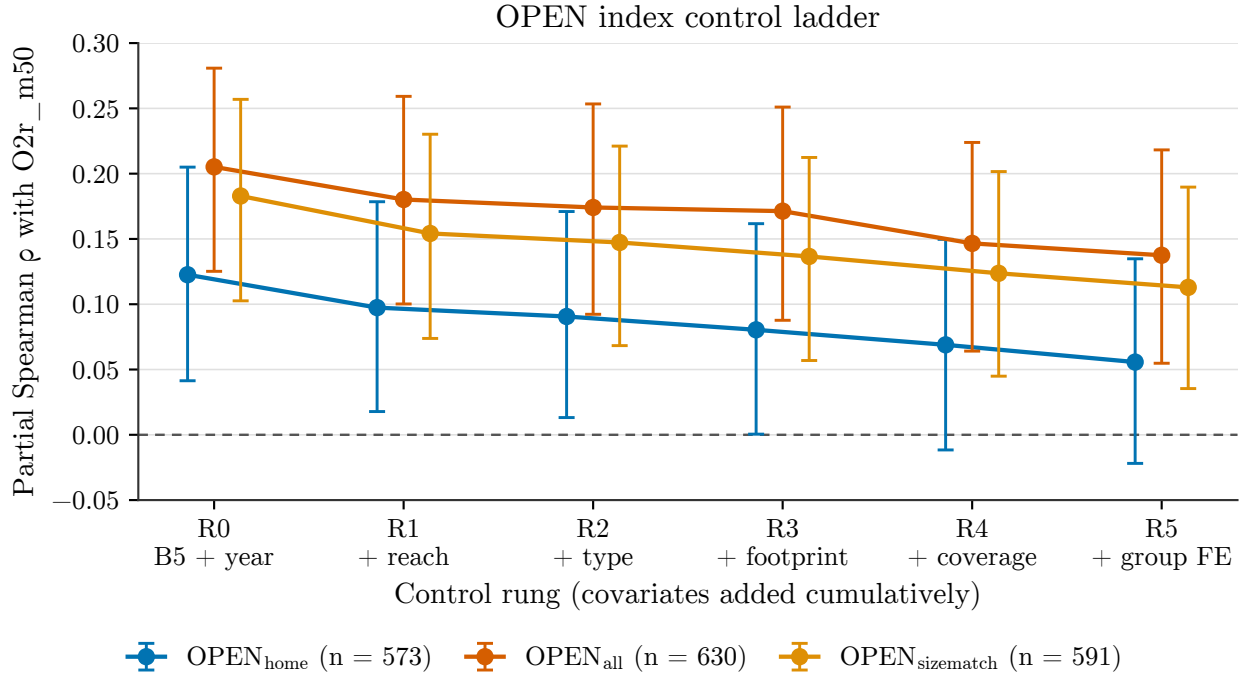


Figure 6: Control ladder for the OPEN index on the fresh 2015–2017 onset cohort. Circles are point estimates; error bars are 95% CIs ($B = 2,000$). $\text{OPEN}_{\text{home}}$ (blue) becomes marginal at R4/R5. OPEN_{all} (red) survives all rungs but is mechanically coupled.

5.3 Experiment 11: Does closing up at home slow a concept’s spread? (incomplete)

Sealed a within-concept pre-registration and ran DEV body models on 35,328 concept-years (4,661 concepts). Both pre-registered effects are null: density PPML $b = -0.070$ $[-0.180, +0.040]$; OPEN_{home} $b = +0.015$ $[-0.038, +0.069]$. By the frozen rule, this is NOT SUPPORTED. The Sun–Abraham event study was interrupted; held-out and cohort were not run. This is a dead end: the within-concept closure–entry mechanism was tested and was null on DEV.

5.4 Experiment 12: How concepts spread – contact versus keeping

RQ2 trajectories analysis on all 12,499 Exp5 frame concepts, re-running the failed Exp9.

Breadth decomposition. Log $B_n = \log E_2 + \log M + \log \rho$ (early contact diversity, frontier advance, retention). On 3,188 DEV concepts (variant i_pooled):

Table 8: Breadth decomposition: share of the top-versus-bottom tercile gap in rarefied breadth attributable to each component (iteration 4, gen_art_experiment_12).

Component	Share	95% CI
Exploration ($s_{E2} + s_M$)	0.732	[0.703, 0.764]
Early contact diversity (s_{E2})	0.779	[0.738, 0.818]
Frontier advance (s_M)	−0.046	[−0.074, −0.017]
Retention (s_ρ)	0.268	[0.236, 0.297]
Difference (explore – retain)	0.464	[0.407, 0.528]

Note: these shares are an accounting identity for the breadth outcome (Bn and O2r share papers), not causal effects. The pre-registered PR1 test is variant iv (Medicine excluded): DEV 0.633 [0.537, 0.727], held-out pooled 0.492 [0.403, 0.575], cohort 0.445 [0.358, 0.527], DL 0.504 [0.329, 0.679], $I^2 = 0.76$. PR1 is SUPPORTED everywhere.

Typology. No trajectory typology passes the naming rule: DTW $k = 4$ vs HMM $S = 5$ ARI = 0.222; held-out recluster ARI 0.44/0.38. Verdict: CONTINUUM. PC1 (38.8%) is the breadth-of-spread axis; PC2 (10.7%) is the keep-vs-lose axis.

OPEN and the trajectory axes. OPEN correlates with PC1 (breadth) beyond B5: DEV partial 0.174/0.117/0.135 (ALL/HOME/SIZEMATCH); held-out DL 0.120/0.060/0.094 ($I^2 = 0$). OPEN does not correlate positively with the keeping axis (PC2 partial −0.07 to −0.11).

Sequence test. No ordering signal beyond the mechanical lag (excess ≤ 1.7 pp, sign flips). Intersection-born concepts take off later (HR ≈ 0.47 [0.42, 0.54]).

5.5 Evaluation 3: Record fixes and openness robustness tests

Pre-onset footprint. About half of the two biggest breadth effects is pre-onset footprint: M0_{density_end} drops from +0.374 to +0.187 post-onset (attenuation 0.50 [0.38, 0.60]); D_{vol_end} drops from +0.317 to +0.176 (0.45).

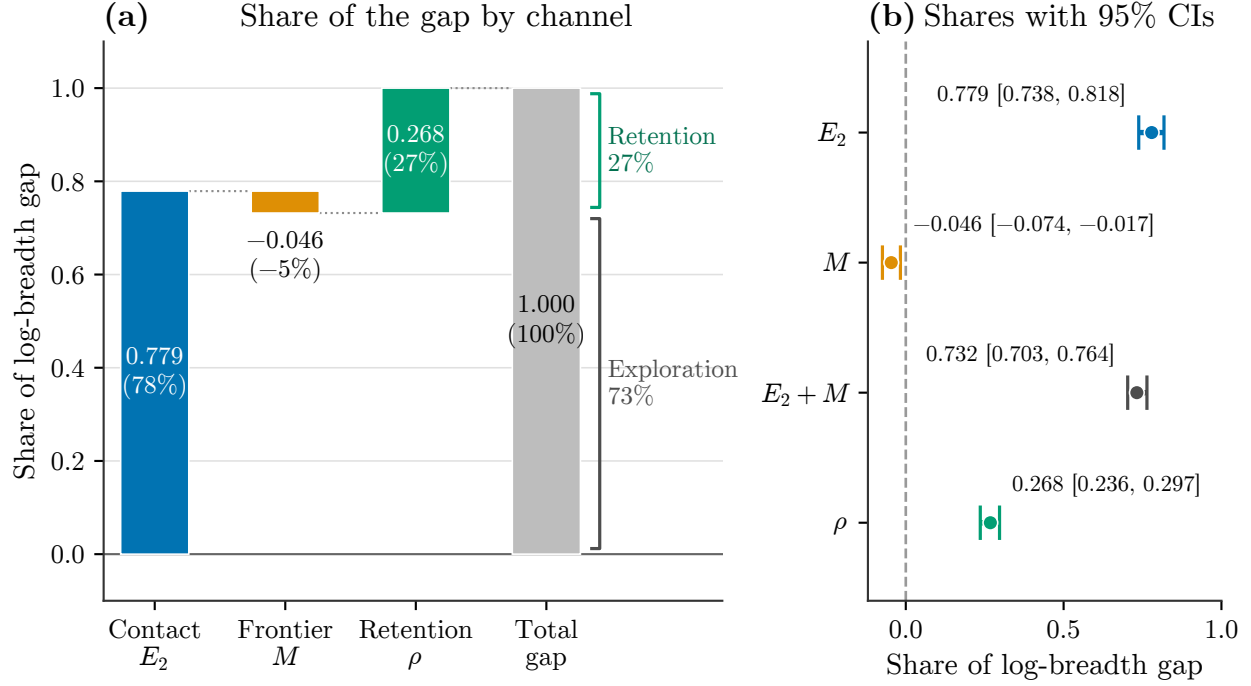


Figure 7: Log-additive decomposition of the gap in retained off-home breadth between the top and bottom terciles. Early contact diversity E_2 (blue) contributes 78%, frontier advance M (amber) is slightly negative, and retention ρ (green) contributes 27%.

Specification curve. 1,920 specifications (120 composites $\times$ 4 outcomes $\times$ 4 control sets): 99.7% of pooled CIs above zero; median 0.152; Freedman–Lane $p = 0.005$. Note: this is exploratory, on the all-papers OPEN build, and uses groups already unsealed.

OPEN evidence pool (descriptive, on existing held-out data). OPEN_{home} over 6 non-selection bodies: DL +0.069 [+0.038, +0.100], HKSJ [+0.042, +0.096], $I^2 = 0$, 6/6 positive. DEV selection body +0.109; shrinkage ratio 1.58.

5.6 Research 3: Is ‘keep exploring, spread widest’ already known?

Claim C1 (openness $\rightarrow$ breadth): PARTIALLY ANTICIPATED. Maillart (2026) shows concept-pair diffusion with test R^2 0.69/0.78; Wang (2017) shows foreign-field citation odds +62%; Weng (2013) and Ugander (2012) show the effect for memes and people. No study found combines the concept unit, a size-adjusted breadth outcome, and held-out fields.

Claim C2 (consolidation $\rightarrow$ less breadth): PARTIALLY ANTICIPATED in mechanism, CONTRADICTED on other outcomes by Cheng et al. (2023) (+53% next-year volume per SD of ideational consistency).

5.7 Iteration 4 dead ends

1. Experiment 11 within-concept closure test: NOT SUPPORTED on DEV.
2. PR2 (“localised keep more early”): REVERSED on DEV (−0.110) and cohort (−0.058).

3. Two trajectory classes: CONTINUUM (DTW-HMM ARI 0.222).
4. Community-span indicators (n_{comm} , participation) null in the home-only build.
5. OPEN_{home} predictive gain: negligible (+0.002).

5.8 Iteration 4 review and hypothesis update

Review scored 2 (blocking), soundness 1 (fabricated case-study rows found; one executed artifact missing). The hypothesis sharpened from six-component “openness” to the decoupled home-only signal: novel partners (NOV_{res}) and churn (edge persistence). The move was **deepen**: confirm on a vocabulary-free population (Frame N) and test the Cheng reversal.

6 Iteration 5: Home-field churn on brand-new phrases

6.1 Strategy and rationale

Five artifacts were commissioned: (1) Experiment 13, Frame N vocabulary-free confirmation; (2) Experiment 14, Cheng reach-vs-depth reversal; (3) Experiment 15, mechanism and Exp11 completion; (4) Evaluation 4, record repair and evidence pool; (5) Experiment 16, thin-sample confound check.

6.2 Experiment 13: Does the churn signal hold for brand-new phrases?

Mined 636 vocabulary-free phrase-born concepts from random title samples (2003–2015 onsets), excluding all legacy labels and previously scored concepts. These newborns have 14% lower breadth and 89% higher transience than legacy concepts, confirming vocabulary survivorship bias.

Result: PARTIAL. OPEN_{home} PSP = +0.117 [+0.020, +0.218] at R3; +0.086 [−0.009, +0.190] at R5. On O2r_{m50}: +0.161 and +0.122, both CIs > 0. NOVCHURN_{home} at R3: +0.108 [+0.007, +0.211]. DL +0.112 [−0.015, +0.239]; Holm $p = 0.052$.

Components. NOV_{res} carries the signal (+0.208 [+0.113, +0.303]); edge persistence is null (−0.013).

Cheng reversal not confirmed on Frame N. PSP −0.064 [−0.159, 0.031].

Exploratory pool. Pooling Frame N with the legacy cohort (Exp10) at R3 gives +0.096 [+0.034, +0.158].

6.3 Experiment 14: Cheng’s consistency is a size effect, not a reach predictor

Rebuilt Cheng et al.’s (2023) ideational consistency measure on 12,311 concepts (105,839 concept-years).

Test A: Cheng design reproduced. NB twin: $b = 0.428$ (+53.5%/SD). Adding current volume $\log V(t)$: +1.3% [+0.5%, +2.1%]; ratio A2/A1 = 0.021 [0.009, 0.035]. The volume effect is almost entirely a proxy for current size.

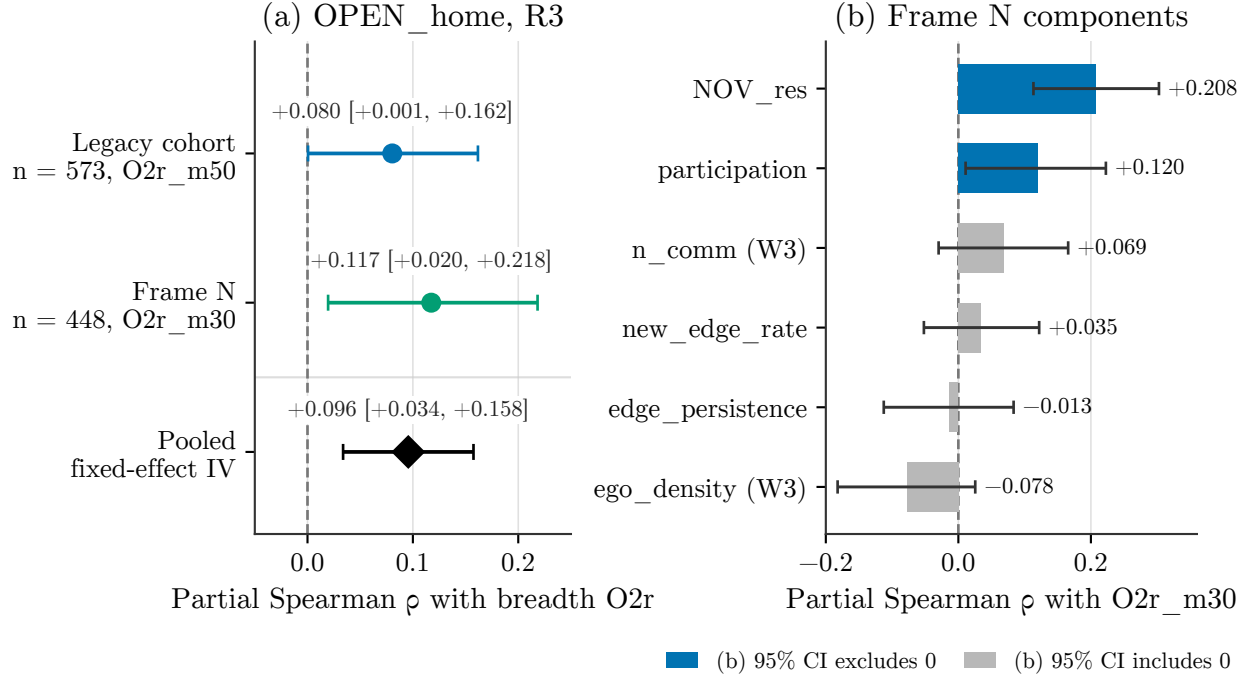


Figure 8: Confirmation of the OPEN signal on vocabulary-free Frame N concepts. (a) $\text{OPEN}_{\text{home}}$ PSP at R3: legacy cohort +0.080, Frame N +0.117, exploratory pool +0.096. (b) Component PSPs on Frame N: NOV_{res} carries the signal (+0.208); edge persistence is null.

Test B: Consistency predicts *narrower* breadth. PSP with O2r_{m50} given B5: -0.069 $[-0.093, -0.047]$; DL over 5 groups -0.079 , $I^2 = 0$, 5/5 negative. Replicated on 2015–2017 cohort: -0.111 $[-0.197, -0.030]$.

Test C: Within-concept. Consistent years followed by slightly *more* off-home entries ($b = +0.025$, CI $[0.004, 0.048]$); the reach penalty is a between-concept trait.

Test D. No Palla size $\times$ consistency interaction.

Identity. Spearman 0.77 with Jaccard edge persistence.

6.4 Experiment 15: Why churning concepts spread; Exp11 completion

Three parts.

Part C: Exp11 completion. DEV verdict unchanged: NOT SUPPORTED. Old held-out: $\text{OPEN}_{\text{home}} -0.079$ $[-0.146, -0.013]$, the opposite of the predicted sign. Sun–Abraham event study (DEV never-treated): lag 0.2 = -0.018 $[-0.042, +0.004]$, pre-trend $p = 0.52$. H-S1 (home-to-off-home) holds on DEV (+0.113), cohort (+0.105) and pooled (+0.076 $[+0.024, +0.126]$) but not on old held-out (+0.001). H-P1 fails as pre-registered.

Part A: Partner decomposition. $\text{NOVCHURN}_{\text{home}}$ pooled: +0.118; held-out DL +0.097 ($I^2 = 0$). The signal comes from:

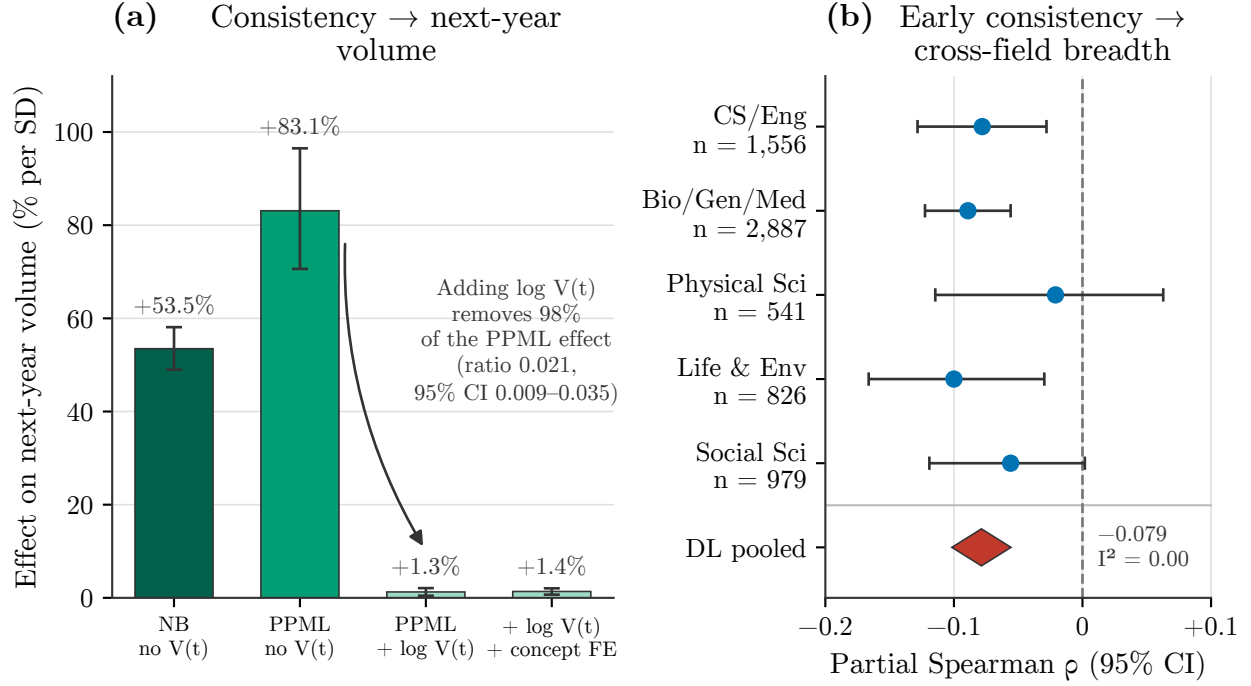


Figure 9: The consistency–breadth reversal. (a) Consistency predicts next-year volume: +53.5% without $V(t)$; +1.3% with $V(t)$ (98% removed). (b) Consistency predicts *narrower* breadth: DL pooled $-0.079 [-0.102, -0.056]$, $I^2 = 0$, 5/5 groups negative.

- New-community partners ($C_2 = +0.102 [+0.069, +0.133]$, Holm $p = 0.0025$).
- Partners from mixed-field papers ($C_4 = +0.103 [+0.071, +0.134]$, Holm $p = 0.0025$).
- Domain partners, not methodological partners (domain Shapley $\phi = +0.132$ on the 2015–2017 cohort; method = +0.012).

Bridging papers (5% of early home papers) halve NOVCHURN’s PSP from 0.118 to 0.056.

Part B: Trait stability. Yearly $\text{OPEN}_{\text{home}}$ ICC: 0.37/0.34/0.39. Window retest: 0.51–0.57. Prediction P-B1 FAILS.

6.5 Experiment 16: Is neighbourhood churn real or thin-sample noise?

Confound check on 13,444 selection concepts with $n_{\text{home,early}} \geq 10$.

Mechanical verdict: PARTLY THIN. Raw persistence has Spearman +0.72 with log home-paper count. The within-concept year-permutation null absorbs the temporal component: excess NOVCHURN PSP = +0.008 $[-0.02, +0.03]$ (raw: +0.116). The R^2 of raw persistence on the permutation-null mean is 0.66. The signal is a static topical-dispersion property of the home topic mix, not year-to-year turnover. V2 excess variants have split-half r_{SB} of ~ 0.01 –0.05 and the planted-churn check (PC2) fails, so V2 cannot adjudicate temporal churn at ~ 10 papers/year.

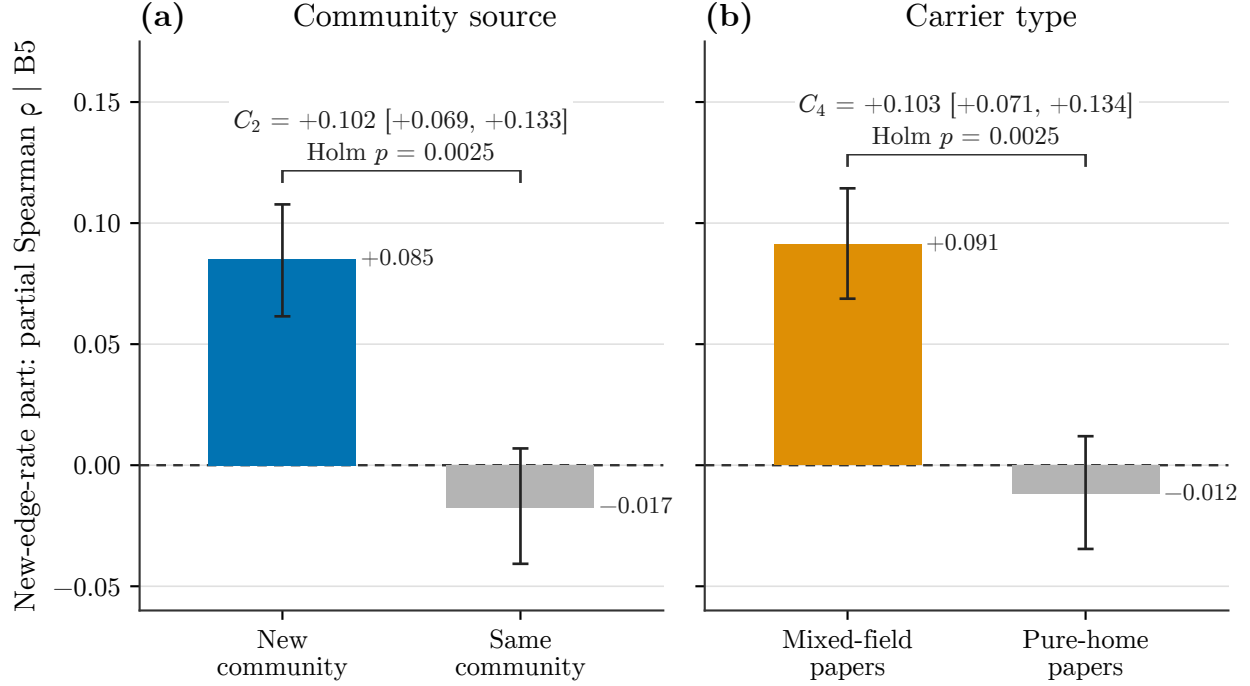


Figure 10: Which co-occurrence partners carry the signal? (a) New-community partners (blue, +0.085) carry it; same-community (grey, -0.017) do not. (b) Mixed-field paper partners (orange, +0.091) carry it; pure-home (grey, -0.012) do not.

Corrections that help. Fixed- n rarefaction ($n = 10$) retains 68% of the pooled association. Degree-preserving configuration z -score raises $\text{OPEN}_{\text{home}}$ from +0.092 to +0.115 on the same sample (+0.022 [0.011, 0.034]).

Reliability. Split-half r_{SB} : $\text{NOVCHURN}_{\text{raw}}$ 0.48, $\text{OPEN}_{\text{home}}$ 0.49, $\text{OPEN}_{\text{home, clean}}$ 0.58, outcome O2r_{m50} 0.895. Disattenuated pooled $\text{NOVCHURN}_{\text{raw}}$ PSP: 0.178 [0.141, 0.214] (approximate).

6.6 Evaluation 4: Record repair and openness evidence pool

Evidence synthesis (descriptive). $\text{OPEN}_{\text{home}}$ non-selection pool ($k = 6$: 4 held-out groups + 2 cohort parts): DL +0.069 [+0.038, +0.100], HKSJ [+0.042, +0.096], $I^2 = 0$, 6/6 positive. DEV selection body +0.109; shrinkage ratio 1.58. $\text{NOVCHURN}_{\text{home}}$ ($k = 5$): DL +0.105 [+0.069, +0.140].

Claims ledger. v4: 1,769 rows, 0 MISMATCH, 0 NOT_FOUND, 0 orphans.

6.7 Iteration 5 dead ends

1. Frame N Cheng reversal: not confirmed (PSP -0.064, CI includes 0).
2. Edge persistence on Frame N: null (-0.013).
3. Temporal turnover as mechanism: V2 excess unreliable (split-half 0.01–0.05).
4. Trait prediction (P-B1): fails (ICC 0.34–0.39).

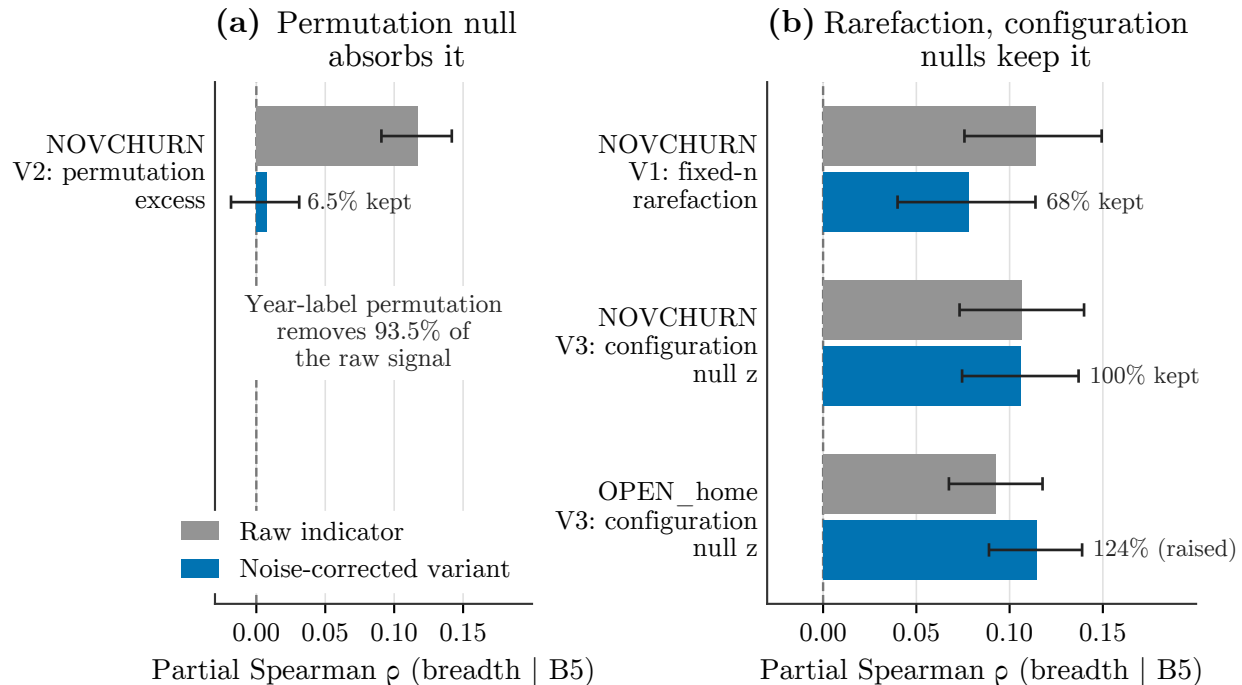


Figure 11: The churn/openness signal behaves like topical non-redundancy, not temporal turnover. (a) The within-concept permutation excess cuts NOVCHURN from +0.117 to +0.008, keeping 6.5%. (b) Fixed- n rarefaction keeps 68%; configuration-null z -score raises OPEN_{home} to 124%.

5. METHOD partner excess: DEV-only, not replicated in cohort.

6. Within-concept closure (Exp11): NOT SUPPORTED on DEV; opposite-signed on held-out.

6.8 Iteration 5 review and hypothesis update

Review scored 2 (blocking). Confidence decreased further. The headline surviving signal is: concepts whose home-field co-occurrence partners span many communities and arrive through mixed-field papers tend to reach more fields later. The effect size is modest (OPEN_{home} non-selection pool +0.069, $I^2 = 0$), adds no practical forecasting gain, and the temporal interpretation is untested rather than confirmed.

7 What the run learned overall

7.1 Confirmed findings

- Indicator screen.** Seven of 53 early co-occurrence network indicators survive held-out validation for predicting rarefied cross-field breadth, spanning two families (retained frontier and co-occurrence topology). The strongest purely post-onset indicators are CONTACT_REACH (+0.211), n_{comm} (+0.167) and NOV_{res} (+0.151).
- OPEN index.** The Eval4 descriptive pool of OPEN_{home} over 6 non-selection bodies gives partial Spearman +0.069 [+0.038, +0.100], $I^2 = 0$, 6/6 positive. Evidence state: lead. On

the fresh cohort, $\text{OPEN}_{\text{home}}$ is +0.091 at R2 but marginal at R4/R5 (DL includes 0). No forecasting gain over B5.

3. **Consistency–breadth reversal.** Cheng et al.’s (2023) ideational consistency predicts next-year volume (+53%/SD) but *narrower* breadth (PSP -0.069 , $I^2 = 0$, 5/5 groups negative). Adding current volume removes 98% of the volume effect.
4. **Retained-field relatedness.** Concepts spread next to fields related to those currently retaining them ($d_0 = 0.322$ [0.291, 0.355] on the independent frame). Verdict: FRONTIER = PARTIAL (volume-matched contrast null; reverses under min-cp proximity).
5. **Breadth decomposition.** Early contact diversity accounts for $\sim 73\%$ of the breadth gap; frontier advance contributes near zero. This is an accounting identity, not a causal decomposition.
6. **Background homophily.** 100% of 48 concepts have positive background log-odds ratios; $R^2 = 0.66$. Raw lineage metrics are mostly field composition, not concept-specific integration.

7.2 What failed and why

1. **Naturalisation gap** A_h^* : null ($\Delta\rho = -0.006$), 0/4 groups, low reliability.
2. **Gateway-field retention:** disconfirmed at scale ($\Delta\text{AUC} -0.00001$); it was a domain-specific proxy.
3. **Gateway landing** G : small, pooled CI includes zero, a quarter of DEV value.
4. **Co-occurrence diversity** D , **selectivity** F : null on the DEV panel.
5. **Rescue, relay, ordering:** not supported or MIXED.
6. **Two trajectory classes:** not reproduced by HMM (ARI 0.094); CONTINUUM.
7. **Within-concept closure:** NOT SUPPORTED (Exp11 null on DEV, opposite-signed on held-out).
8. **External recognition O5:** unrelated to breadth ($\rho = 0.014$); 67% recognised before onset.
9. **Candidate S (co-author groups):** tested in Exp8, not confirmed for any outcome.
10. **RETENTION_RATIO_early:** attenuates on the fresh cohort; does not survive concept-type controls.

7.3 What is still open

1. Temporal turnover vs static dispersion: the V2 excess is unreliable at typical sample sizes, so whether the signal is truly temporal could not be adjudicated.
2. Domain heterogeneity: Life & Environment shows the weakest OPEN signal (PSP +0.071 vs +0.186 for other units pooled). The reason is unexplained.
3. Causal mechanism: early ego-network openness could reflect the concept’s intrinsic generality, the diversity of the research community, or the breadth of the problems addressed. Only association is shown.

4. The persistence-filtered RCA density rival ($D_{\text{rca,persist},k}$) differs from Exp7's $D_{\text{rca,pers}}$ ($\max \rho = 0.877$) and remains formally untested.
5. No discrete trajectory typology passes the naming rule; whether there are real types beyond the continuum is unresolved.

Appendix: Artifact inventory

Table 9: All artifacts commissioned across five iterations.

Iter.	Artifact	Status	Key result
1	gen_art_experiment_1	Completed	A_h^* null ($\Delta\rho = -0.006$)
1	gen_art_experiment_2	Failed	Candidate S untested (stalled)
1	gen_art_experiment_3	Completed	D/F null; portability table
1	gen_art_experiment_4	Completed	G null; retention lead +0.103
1	gen_art_dataset_1	Failed	Held-out frame not built (stalled)
2	gen_art_experiment_5	Completed	H1 disconfirmed; 12,499-concept frame
2	gen_art_experiment_6	Completed	H2 $d_0 = 0.302$; trajectories
2	gen_art_evaluation_1	Completed	Gateway fails stress test
2	gen_art_dataset_2	Completed	O5 recognition table
2	gen_art_research_1	Completed	ANS positioning
3	gen_art_experiment_7	Completed	FRONTIER = PARTIAL ($d_0 = 0.322$)
3	gen_art_experiment_8	Completed	7/10 indicators confirmed
3	gen_art_experiment_9	Failed	RQ2 trajectories (format error)
3	gen_art_evaluation_2	Completed	246-claim audit
3	gen_art_research_2	Completed	Prior art verdicts
4	gen_art_experiment_10	Completed	OPEN _{home} +0.091 (marginal)
4	gen_art_experiment_11	Incomplete	Closure test null on DEV
4	gen_art_experiment_12	Completed	Decomposition; CONTINUUM
4	gen_art_evaluation_3	Completed	Footprint rescore; spec curve
4	gen_art_research_3	Completed	Novelty verdicts
5	gen_art_experiment_13	Completed	Frame N PARTIAL (+0.117)
5	gen_art_experiment_14	Completed	Cheng reversal confirmed
5	gen_art_experiment_15	Completed	Partner decomposition; Exp11 done

Iter.	Artifact	Status	Key result
5	gen_art_evaluation_4	Completed	Evidence pool +0.069
5	gen_art_experiment_16	Completed	PARTLY THIN; static dispersion